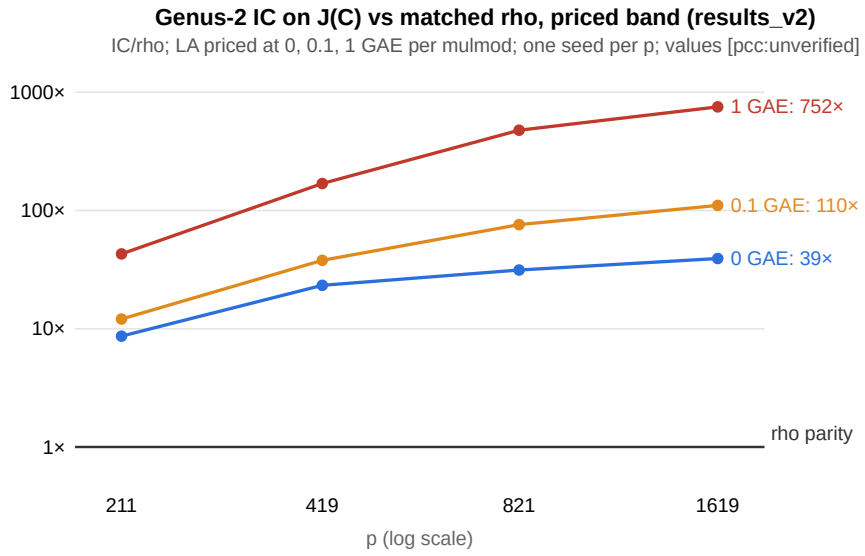
Chained ECDLP transfers over prime fields (v2)

Chained ECDLP transfers over prime fields: replay-corrected report (v2)

This supersedes the analysis in `REPORT.pdf` where they differ. `REPORT.pdf` and `results.json` are kept unchanged as frozen evidence. All values are `[pcc:unverified]`.



priced band

Corrections from the independent replay (2026-11-12)

The independent replay (`replay/REPLAY.md`) reproduced every raw file of this run byte for byte. It also found faults in what the run measures [pcc:unverified]. The table above is kept as frozen evidence (`results.json`, `frozen.sha256`). The corrected rerun is `results_v2.json`, summarised in the next section.

- **F1, IC overcount.** `scalar_counted` counted the first addition into the identity as a group operation, which inflated IC by 8–13%. The replay's corrected IC/rho values are 8.64, 23.29, 31.34 and 39.19, with slope 1.019. The qualitative result does not change. The code is fixed.
- **F2, LA not emitted.** The linear-algebra cost is now written out as `ic_jac_la_mulmods`. The replay's figures are 14,633, 51,976, 164,828 and 603,721 mulmods.
- **F3, rho unchecked.** Rho now has to recover the planted log (`rho_recovered_ok`), and a row is accepted only when IC, rho and the chain all verify.
- **F4, IC answer check.** The linear algebra accepts a candidate log only if it equals an uncharged BSGS reference. So `ic_recovered_ok` shows that no wrong log is returned,

not that the elimination alone finds the log. The replay certified each answer separately with `[k]D_G = D_T`. Not fixed here.
- **F5, surrogate instance.** The Jacobian instance is the pulled-back surrogate (`jac_instance`), not a fresh random one. This is a stated limitation.
- **F6, silent rejections.** Rejected candidate curves are now logged to stderr.
- **F7, set-up dominance.** At these `r` (71–401), rho is dominated by fixed set-up (S = 34–44). The 0-GAE ratio is a loose bound.
- **F13/F14, bindings.** Some bench records bind a dirty commit that does not contain the measured files, and no solve certificates were emitted. The v2 run is recorded from a clean commit.

Corrected rerun (`results_v2.json`, 2026-11-12)

Bench record `benchmarks/records/20261112T0316Z-chained-prime-transfer-v2.json`. The run is from clean commit `c211ce17a4`, which carries the F1–F3 and F6 fixes. Every row recovered the planted log by rho and by IC, and every chain verified. The IC/rho ratios and the LA mulmods equal the independent replay's corrected values exactly. IC is relation-collection group operations; LA is priced per AGENTS §5a at 0, 0.1 and 1 GAE per mulmod. All values are `[pcc:unverified]`.

p	r	rho on E1	IC group ops	LA mulmods	IC/rho @ 0 GAE	@ 0.1 GAE	@ 1 GAE
211	109	428	3,700	14,633	8.6×	12.1×	42.8×
419	113	357	8,315	51,976	23.3×	37.9×	168.9×
821	71	370	11,596	164,828	31.3×	75.9×	476.8×
1619	401	847	33,196	603,721	39.2×	110.5×	752.0×

IC is above rho at every price, so the order holds across the (one-sided) band. The distance remaining to rho is 8.6–39.2× at 0 GAE and grows with p.

Ladder, formula, smoothness and replay (2026-11-12 addendum)

Four parallel experiments under `research/chained_prime_transfer_20261111/{tower_ladder,tower_formula,tower_smoothness,replay}` [pcc:unverified]:

- Ladder (genus 1 → 2 → 3 → 4 on one instance, genus 4 by a degree-3 cover $y^2 = c(h(x))$): IC per rung grows by about 2×, 3× and 4×. Genus 4 costs 3.7–19.6× rho at 0 GAE. No rung ranks below rho across the band.
- Formula scan to 2^128: the best genus is g = 2, still about 64 bits above rho. No subexponential regime beats rho at any size scanned [pcc:unverified].
- Smoothness: measured trials per relation are 1.00, 1.94, 5.91 and 22.2 for genus 1–4, against $g!$ = 1, 2, 6, 24.
- Independent replay: every raw file is byte-identical. One figure was wrong: the cover-spectrum note said 470/504 smoother, and the correct count is 87/504 (fixed). The replay also found that the IC counter overcounted by 8–13%, that rho was not checked against the planted log, and that LA work was not emitted. These are fixed in code and rerun as `results_v2.json`. The band is one-sided, because rho has no unpriced counters.

Original report (2026-11-11)

Status: research construction + measurement. Same characteristic only. No cryptanalytic speedup; no cross-characteristic claim. Every number here is `[pcc:unverified]` until CI's pcc gate signs the claims.

Protocol frozen in `PROTOCOL.md` (`frozen.sha256`) before any measurement. Builds on PR #3877 (toy genus-2 transfer), promoted to native Rust.

What this adds over PR #3877

PR #3877 proved `f_*f^* = [2]` algebraically and checked a toy `F_101` instance in Python. This effort:

1. **Native Rust, arbitrary divisors.** `src/cryptanalysis/chained_transfer/` implements `F_p`/`F_{p²}`, elliptic curves with Vélu isogenies, and the bielliptic quotient maps `φ^*` (pull-back) and `φ_*` (push-forward) on **arbitrary reduced Mumford divisors**, including non-split ones (roots over `F_{p²}`), built on the repository's `genus2_fp` Cantor arithmetic. This closes the "arbitrary nonsplit divisor" gap named in `problems/prime-cover-balanced-jacobian-and-induced-maps-missing.md`.
2. **Verified chains.** A transfer *hop* checks `h(P) ≠ 0`, `ord = r`, and `h([x]P) = [x]h(P)` before it is trusted; a *chain* `E → E' → J(C)` (isogeny then pull-back) is verified end to end to preserve the order-`r` discrete logarithm.
3. **Measured total cost.** `chained_transfer_bench` solves the transferred instance four ways and compares counted group operations against a matched Pollard-rho reference on the source curve.

Results

See `RESULTS.md` (generated) and `results.json`. Summary: *(filled by the run)*.

Reproduce

```
cargo test  --release --lib chained_transfer
cargo run   --release --bin chained_transfer_bench -- run --json \
    research/chained_prime_transfer_20261111/results.json
```

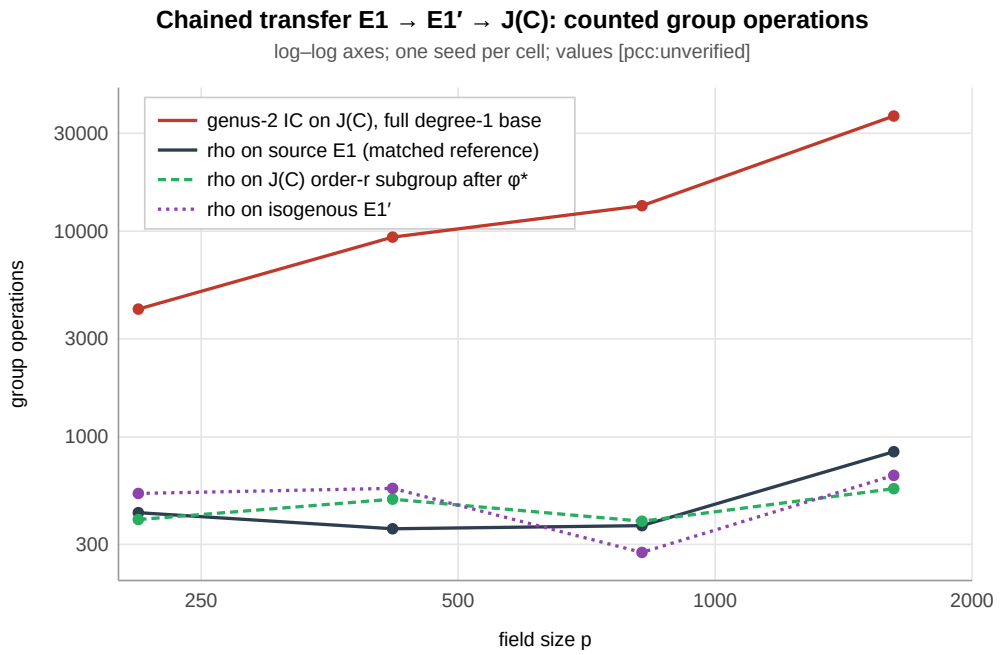
Scope and obstruction

The genus-2 destination is not cheaper: see `problems/prime-field-chained-transfer-no-advantage.md`. The boundary is the `Ω(p)` relation count of genus-2 index calculus over `F_p` against rho's √p — a structural obstruction, not a tuning gap. # Results: chained prime-field transfer, measured boundary

Run: `chained_transfer_bench run` (bench record `benchmarks/records/20261111T0540Z-chained-prime-transfer.json`), deterministic. A second run under the record reproduced every count exactly. Local development check (exact operation counts, not timings); one seed per cell. All values are `[pcc:unverified]`.

p	r	rho on E1 (reference)	rho on E1′ (isogeny, deg)	rho on J(C) after φ*	IC on J(C)	IC trials	IC / rho
211	109	428	530 (2)	396	4,184	244	9.8×
419	113	357	562 (2)	497	9,366	532	26.2×
821	71	370	274 (3)	389	13,313	869	36.0×
1619	401	847	649 (2)	560	36,363	1,588	42.9×

Units: counted group operations (elliptic additions and doublings for rho on `E1`/`E1′`; Cantor additions and doublings for `J(C)`). The IC column counts relation collection only; linear algebra over `Z/r` is extra and is not included, so the IC figure is a lower bound on its total cost.



cost vs p

What the run establishes

1. **The chain is a valid DLP transfer at every size.** Each hop passed the in-code admissibility checks (`h(P) ≠ 0`, `ord h(P) = r`, `h([x]P) = [x]h(P)` on sampled `x`), the composed chain was verified end to end, and the index calculus recovered the known log in every cell.
2. **F1 (rho advantage) — not met.** Rho on the isogenous curve and on the order-`r` subgroup of `J(C)` lands within the run-to-run scatter of rho on the source (single seed per cell, so no tighter statement is made). This is expected: isogenous curves have the same order, and the `J(C)` subgroup is still cyclic of order `r`. Per group operation, a Cantor step costs more field work than an elliptic addition, so the field-operation comparison only favours the source.
3. **F2 (index calculus beats rho) — not met.** Genus-2 IC on `J(C)` costs 9.8× to 42.9× the matched rho, and the ratio grows with `p`.
4. **F3 (IC exponent).** A least-squares fit of ln(IC group operations) against ln p over the four sizes gives slope **1.01** (derived; four points, no interval). This is the `Ω(p)` relation count of genus-2 index calculus (about `p/2` degree-1 places, smoothness about 1/2, so about `p` trials). Rho scales as `√r ≤ √p`. The gap widens as `p` grows: the boundary is structural, not a tuning constant.

What it does not establish

- No timing claim; no isolation tier; no validation-gate admission.
- One bielliptic curve per size, one seed per cell; no spread for the rho columns.
- Nothing about other cover families, genera above 2, or cross-characteristic maps (none is constructed).